

TI-99 Console Tester Public Beta 0.8

This is the hardware-validated public beta for the 16 KiB W27C512 modification of Geoff Trott's TI-99/4 and TI-99/4A Console Tester.

Which file to program

Program `HEXDIAG08.BIN` into the W27C512 used by the tested 16 KiB modification. It is the complete 65,536-byte image with the extension at `>6000->7FFF`, the fixed core at `>E000->FFFF`, and unused space filled with `>FF`.

Use `HEX8K08.BIN` for the first electrical/compatibility check. It repeats the fixed core in all eight physical W27C512 banks so A13, A14, and A15 cannot select different contents.

`HEXCORE8.BIN` and `HEXEXT08.BIN` are the two raw 8 KiB component images for developers. They are not complete W27C512 programmer files.

Before installation

1. Read `MANUAL08.PDF` and `HARDWARE.PDF`.
2. Compare the selected file with `SHA256.TXT`.
3. Program the entire device and run the programmer's verify pass.
4. Check the modification with the board unpowered.
5. Begin with the repeated-core image if the decoder or wiring is unproven.

The VRAM and memory tests are destructive. Do not run the diagnostic with unsaved machine state.

What changed in Beta 0.8

Beta 0.8 identifies TMS-family, F18A, Pico9918, and other F18A-compatible video interfaces; keeps both joystick ports visible while giving a clear Alpha-Lock advisory; and fully restores VDP and sprite-table state after every visual test. It also adds strict AORG/XORG collision checking and protects the full font and keyboard tables against accidental overlays. See `CHANGES08.PDF` for details.

The sprite page contains five entries but normally shows three groups at the original four-sprites-per-line limit: the overlapping red/white collision pair, cyan, and light blue. Yellow appears as a fourth group only when an enhanced VDP is configured above that limit. This is not a Pico9918 problem and is not a pass/fail condition.

Qualification

The `HEXDIAG08.BIN` bytes passed the complete static acceptance suite and were exercised on a stock TI-99/4A, TI-99/4QI V2.2, a GROMless TI-99/4A through the LOAD recovery path, and a TI-99/4 fitted with a Pico9918. Observed video interfaces included TMS family, F18A, and Pico9918.

License and credit

Source and build tooling are Apache-2.0. Original documentation and illustrated modification material are CC BY 4.0. The attributed MegaDemo-derived material has the separate permission boundary described in the repository's `docs/PROVENANCE.md`. The enhanced-VDP detector adapts code supplied by Troy Schrapel under the MIT License; the complete notice is in `TROY-MIT.TXT`.

Credit **Jon Guidry (hexbus)** and link to the original project: github.com/hexbus/ti99-console-tester.